

Design and simulation of a broadband tunable transmittance optical filter using Sb₂Se₃ phase change material

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Abstract: We propose a switchable transmittance filter that operate around 1064 nm wavelengths using antimony triselenide (Sb₂Se₃). Sb₂Se₃ is a phase change material (PCM) that has low absorption in the infrared (IR) spectrum, large and stable refractive index contrast between the two phases, and high optical endurance in an order of 10⁶ cycles. We provide the reasoning for selecting a long-pass edge-filter design and compare its performance to other planar thin-film designs. At 1064-nm, the selected 5-unit-cells edge-filter design has 93.6% transmittance in amorphous phase and 0.2% in crystalline phase, a 93.4% transmittance contrast with ~200 nm FWHM. It has minimal transmittance degradation after enduring over 1.5×10⁶ phase switching cycles. We show the transmittance tunability of the design by changing the phase of each Sb₂Se₃ layer, and each layer thickness can be optimized to increase the optical performance across the 900-1200 nm wavelengths.

1. Introduction

The term phase change material (PCM) can have different meanings depending on the research field. In energy storage, the term refers to material that store and release thermal energy through changes in its state of matter (latent heat), such as between solid and liquid states [1]. However, in electronics and photonics, the term phase change material refers to a solid-state material that changes its crystal phase when its temperature rises above the transition temperature (T_{trans}). This change leads to a contrast of complex refractive index ($\tilde{n}$) between the initial and post-transitioned phases, where $\tilde{n}$ comprises of the real refractive index (n) and the imaginary extinction coefficient (k) parts [2]. The refractive index contrast between the two phases is an order of magnitude larger than the contrast caused by an applied electric field or light intensity (Kerr effect) [3], which makes PCM ideal for photonic applications that require the modulation of optical performance. Examples include reconfigurable metasurfaces [2, 4-6], optical limiters [3, 7-9], optical switches [10, 11], optical filters [12-15], and absorbers [16, 17].

Phase change material can be classified based on its behavior after phase transition. A volatile PCM reverts to its pre-transitioned phase when the temperature falls below T_{trans} . However, a non-volatile PCM will transition for amorphous to crystalline phase and remains in crystalline phase even if the temperature falls below T_{trans} . To reverts its phase back, it needs to be melted and then rapidly quenched into amorphous phase, the quenching speed needs to be fast enough such that PCM atoms don't have time to for crystalline lattice. An example of volatile PCM is vanadium dioxide (VO₂), which in the past 10 years has gathered attention as a volatile PCM with T_{trans} closest to room temperature at 68 °C. It also has a relatively large refractive index contrast starting from near-infrared (NIR) with a small extinction coefficient (k) in the insulating phase (cold state), however, k increases when transitioned to the metallic phase (hot state) [8, 9]. Two examples of non-volatile PCMs are chalcogenide glasses,

$\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) and $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}$ (GSST). GST was one of the earliest and most studied PCM [18]. It has a large refractive index contrast between amorphous and crystalline phases, low T_{trans} when compared to other non-volatile PCM, $T_{\text{trans}} \sim 145^\circ\text{C}$ for the cubical phase, and $T_{\text{trans}} \sim 200^\circ\text{C}$ for the hexagonal phase [3, 19]. A drawback is its large extinction coefficient in both the visible and infrared (IR) spectra, which causes high optical loss from absorption. Meanwhile, GSST ($\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$), which is a derivative of GST that substitutes some part of tellurium with selenium, also has a large refractive index contrast but a lower extinction coefficient, which helps increase the optical efficiency. However, GSST has a higher T_{trans} of $\sim 242^\circ\text{C}$ [19].

Antimony triselenide (Sb_2Se_3) initial research is in solar-energy applications, it is a relatively new PCM in the field of photonics with non-volatile property, near-zero absorption in NIR and IR wavelengths, large and stable refractive index contrast from visible to IR, and a transition temperature that is between GST and GSST at $\sim 186^\circ\text{C}$ [20, 21]. In addition, Sb_2Se_3 has a fast phase-switching speed in the microsecond range [18, 21, 22], and its phase-switching endurance is also high, with experimental demonstrations showing that it can reach as high as 10^6 to 10^7 cycles [21, 22]. These properties make Sb_2Se_3 an attractive material for high-efficient reconfigurable optical switches and optical circuits that can be used in optical waveguide [4-6, 11, 14, 18], or an optical filter [14, 15].

The transmittance of optical filter can be modulated by the changes in refractive index of PCM. The ON mode has high transmission by creating antireflection condition and minimizing the absorption from the filter, and OFF mode has low transmission after phase transitioned. The methods for phase transition can be via absorption of high-intensity laser or from direct heating of PCM layers (Joule heating) [12]. The former method is similar to optical limiters' but is not suitable for Sb_2Se_3 since it has very small extinction coefficients in both phase at IR wavelengths. So direct heating or heating via a different wavelength light, such as visible spectrum [21], is preferred. The designs can be planar thin-film structures or metasurfaces; planar structures are easier to design, optimize, and fabricate, while metasurfaces offer higher design flexibility [8]. For planar designs, there are many methods for creating antireflection condition, such as single-layer or multilayer antireflection designs, line-pass filters, or Fabry-Pérot resonant cavities [23].

In this work, we investigated Sb_2Se_3 as a PCM in planar thin-film designs for transmittance tunable optical filters. The operating wavelength was selected to be 1064 nm, Nd:YAG laser wavelength which is commonly used in laboratory setups. At this wavelength, Sb_2Se_3 exhibits zero absorption in the amorphous phase and near-zero absorption in the crystalline phase, and its refractive index contrast is high and stable throughout the IR spectrum. The goal is to create a low-loss, broadband transmittance-tunable optical filter.

2. Theories

2.1 Optical properties of Sb_2Se_3

Antimony triselenide (Sb_2Se_3) is an orthorhombic chalcogenide non-volatile phase change material (PCM) characterized by quasi-one-dimensional ribbons. While its amorphous phase consists of a disordered network of distorted motifs, crystallization restores long-range order along the ribbon direction. This thermally driven transition can be triggered by nanosecond to microsecond optical or electrical pulses [18, 21], the process is reversible and has high endurance, with optical switching reaching 10^6 to 10^7 cycles [21, 22]. As an optical device, Sb_2Se_3 enables repeatable toggling between high and low transmission modes, provided that local heating remains below the melting or ablation threshold [4, 22].

The complex refractive index ($\tilde{n}$) of any phase change material at wavelength λ is:

$$\tilde{n}_p(\lambda) = n_p(\lambda) + ik_p(\lambda) \quad (1)$$

where the real term n is the refractive index, the imaginary term k is the extinction coefficient, and p denotes the phase of Sb_2Se_3 which can be either amorphous (*amo*) or crystalline (*cry*) phase. Dispersive models, such as Tauc-Lorentz or multi-oscillator fits are often used to represent the measured n_p and k_p over the spectral range of interest. However, in this work, linear interpolation is used because all the datasets used have measurement interval of less than 4 nm. Figure 1 shows their complex refractive indices from 300 to 1700 nm using the dataset from Delaney, Zeimpekis, Lawson, Hewak and Muskens [18], where the complex refractive indices at 1064 nm are $\tilde{n}_{amo} = 3.370 + 0.000i$ and $\tilde{n}_{cry} = 4.308 + 0.010i$. The blue lines represent real refractive index (n) and the red lines represent imaginary extinction coefficient (k). In amorphous phase (Fig. 1a), the refractive index increases from ~ 2.4 at 300 nm to a peak of ~ 4.0 at 600 nm and gradually decreases to a stable value of ~ 3.3 above 1200 nm, indicating a strong dispersion near the band edge of visible spectrum and a weaker dispersion in the NIR spectrum. The extinction coefficient starts from ~ 2.1 at 300 nm and drops to zero at ~ 850 nm. In the crystalline phase (Fig. 1b), the refractive index at the UV wavelengths is around the same value as the amorphous phase but reaches a higher peak of ~ 5.1 at around 660 nm; it then drops to a stable of ~ 4.1 in the NIR spectrum. The k -index of crystalline phase peaked at ~ 3.1 in the near-UV spectrum and rapidly approached zero above 900 nm.

The refractive index contrasts ($\Delta n = n_c - n_a$) are large and stable from the visible through IR spectra, and the extinction coefficients are large in the near-UV region but fall to zero around the NIR spectrum for both phases, with the crystalline phase exhibiting higher absorption in the visible spectrum. Thus, when compared to other PCMs, Sb_2Se_3 has several advantages when operating in the NIR and IR wavelengths. The large Δn and low k starting from NIR wavelengths enable a device to operate in two modes with little to no optical loss, and a stable Δn also helps with broadband designs.

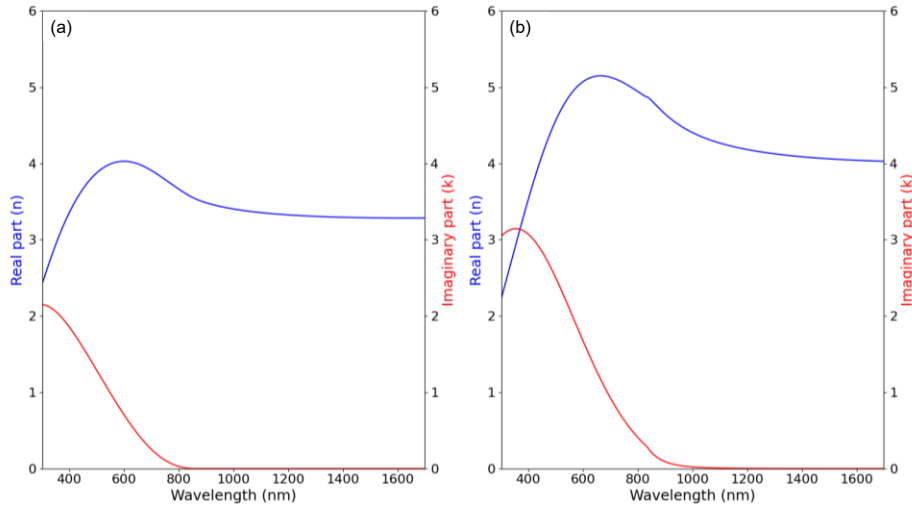


Fig. 1. Complex refractive index of Sb_2Se_3 in (a) amorphous, and (b) crystalline phases. Graphs are plotted from experimental data of Delaney, Zeimpekis, Lawson, Hewak and Muskens [18].

2.2 Single-layer antireflection design

The goal of designing most optical devices is to either create antireflection or high-reflection conditions. For two-mode optical device, such as transmittance tunable filter, we can consider the reflection in the ON mode first, and then the reflectance contrast (ΔR) between the ON and OFF modes. The simplest antireflection from a planar thin-film design is one film layer deposited on top of a substrate; both the film and the substrate should have zero absorption.

The diagram in the top right corner of Fig. 2 shows that a single-layer film reflects light at two interfaces: the bottom interface between the film and substrate (R_1) and the top interface between the air and film (R_2). Antireflection occurs from the destructive interference between R_1 and R_2 , with a complete AR condition requiring the same amplitude and a phase difference of π radian (optical path length of $\lambda/2$). Because R_1 propagates through the thin film layer twice (while transmitted down and reflected backup), the optical thickness (OT) of the film only needs to be equal to half of the optical path length, $OT = \lambda/4$, this also known as quarter-wave optical thickness (QWOT). The optical thickness is related to the physical thickness (t) by the following equation:

$$OT(\lambda) = n(\lambda)t, \quad (2)$$

where $n(\lambda)$ is the refractive index of the film at λ . For a single-layer AR device, the thin-film physical thickness (t) must be:

$$t = \frac{OT}{n} = \frac{\lambda/4}{n}. \quad (3)$$

Another method for calculating the thickness required for the AR condition is to consider the reflection of the device. The reflectance (R) of only the substrate is related to the Fresnel coefficient (r) by:

$$R = |r|^2 = \left| \frac{n_a - n_s}{n_a + n_s} \right|^2, \quad (4)$$

where n_a is the air refractive index, and n_s is the refractive index of substrate. If a single-layer film is placed on top of a substrate, the film and substrate layers can be combined into an effective reflectance index (n_r) [23], defined as:

$$n_r = n_f \frac{(n_s + n_f) + (n_s - n_f)e^{-i2\theta}}{(n_s + n_f) - (n_s - n_f)e^{-i2\theta}}, \quad (5)$$

where n_f is the film refractive index, i is the imaginary unit, and θ is the phase thickness which is related to optical thickness (OT) and physical thickness (t) by the angular wavenumber ($k_\theta = 2\pi/\lambda$):

$$\theta = k_\theta OT = k_\theta nt. \quad (6)$$

By replacing n_s in Eq. 4 with n_r from Eq. 5, the reflectance of a single-layer device becomes:

$$R = |r|^2 = \left| \frac{n_a - n_r}{n_a + n_r} \right|^2. \quad (7)$$

Antireflection condition happens when $R = 0$ or $n_r = n_a$. This condition occurs when the phase thickness (θ) is in multiple of $\pi/2$ radian:

$$\theta = Z \frac{\pi}{2}, \quad (8)$$

where Z denotes even integer. If this condition is satisfied, Eq. 5 can be simplified with Euler's identity ($\exp^{-i\pi} = -1$) to:

$$n_r = n_f \frac{(n_s + n_f) - (n_s - n_f)}{(n_s + n_f) + (n_s - n_f)} = \frac{n_f^2}{n_s}. \quad (9)$$

By substituting this into Eq. 7, the required refractive index for antireflection condition is:

$$n_a n_s = n_f^2. \quad (10)$$

Another condition is the relative refractive index between the air, film, and substrate, which must be in the following order: $n_{air} < n_f < n_{substrate}$. Light shifts its phase by π radian when reflected from a lower to a higher refractive index medium. The reflected light at the air-film interface (R_2) will always undergo a π radian phase shift since n_a will be the lowest in most cases. Therefore, R_1 must undergo the same phase shift to create destructive interference.

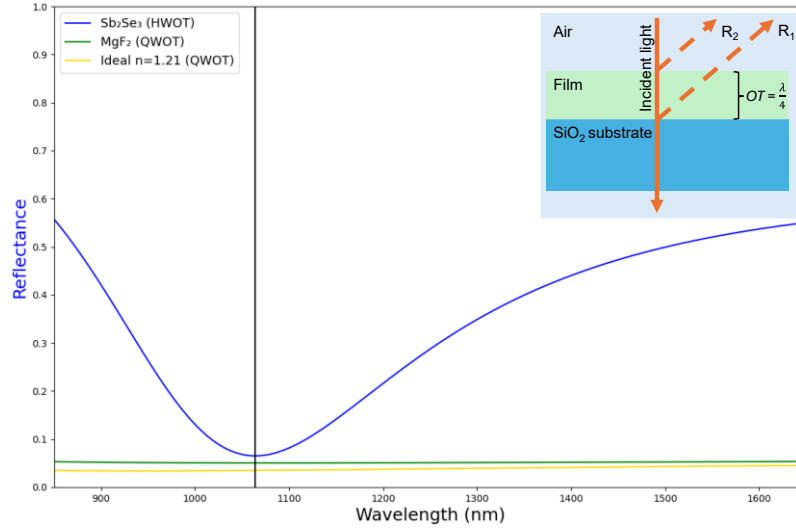


Fig. 2. Reflectance comparison of a single-layer design with different film. At 1064 nm (black, vertical line), HWOT-thick Sb_2Se_3 (blue) has the lowest reflectance $R = 6.5\%$. QWOT-thick MgF_2 (green) $R = 5.0\%$, slightly higher than an ideal film with $n_{\text{ideal}} = 1.21$ (grey) $R = 3.4\%$. The QWOT-thick structure is shown in the top right corner.

These two conditions limit material selections for a perfect antireflection, since there is only one ideal refractive index (n_{ideal}) for each n_s . The refractive index of Sb_2Se_3 is higher than that of substrate (silica; SiO_2) at every wavelength. So, if Sb_2Se_3 is QWOT thick, the reflection from R_1 still travels 2QWOT (π radian) further than R_2 but only R_2 has its phase shifted by π radian during reflection, so the two Rs have a phase difference of 0 radian, a constructive interference. To minimize reflection, the Sb_2Se_3 thickness can be increased to twice the QWOT ($OT = \lambda/2$), this is called half-wave optical thickness (HWOT) and is equivalent to an absentee layer. For example, at the reference wavelength $\lambda_0 = 1064$ nm, if we use 3-mm-thick SiO_2 as the substrate ($n_s = 1.46$) [24], the ideal QWOT-film thickness is calculated from Eq.10 to be $n_{\text{ideal}} = 1.21$. The material with the closest refractive index is MgF_2 ($n_{\text{MgF}_2} = 1.38$) [25]. A slight difference from ideal value causes the reflection to be slightly higher, as illustrated in Fig. 2. An ideal film still exhibits small reflectance because of the reflection in the substrate-air layer in the back. The reflectance of the Sb_2Se_3 (blue) is the highest because the PCM layer is an absentee layer, so the reflection is equivalent to a standalone 3-mm SiO_2 substrate.

2.3 Multilayer antireflection design

Two-layer design with each layer thickness being QWOT (Q/Q design) is also be used to create antireflection conditions. The Fresnel reflection coefficient (r) from Eq. 7 can be expanded to:

$$r = \frac{n_a - n_{r2}}{n_a + n_{r2}} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}, \quad (11)$$

where n_f begins from the substrate side and increases toward the air side ($n_s/n_{f1}/n_{f2}/n_a$), n_{r2} is the effective reflectance index of the entire structure [23], and n_{r1} is the effective reflectance index of only the substrate (n_s) and bottom film (n_{f1}). Similar to Eq.7, the antireflection condition of the two-layer design occurs when $n_a = n_{r2}$ or:

$$n_a = \frac{n_{f2}^2}{n_{f1}/n_s}, \quad (12)$$

$$\frac{n_a}{n_s} = \frac{n_{f2}^2}{n_{f1}^2}. \quad (13)$$

From Eq. 13, it can be seen that the refractive index of the top film must be smaller than the bottom film ($n_{f2} < n_{f1}$). For example, at $\lambda_0 = 1064$ nm, if we use the same SiO₂ substrate ($n_s = 1.46$) and Sb₂Se₃ as the bottom film ($n_{f1} = 3.37$), the ideal refractive index of the top film (n_{f2}) is 2.78. A commonly used material with the closest refractive index is TiO₂ ($n_{TiO2} = 2.31$) [26]. Compared to the single-layer Sb₂Se₃ design, the two-layer Sb₂Se₃/TiO₂ design has a lower reflectance at 1064 nm and wider AR region, as shown in Fig. 3. Another way to create a two-layer AR condition is to fix the two film refractive indices and vary the thickness of each film, as explained in Chapter 5.2 of [23]. However, this method narrowed the antireflection region.

A three-layer design with an HWOT-thick film in the middle (Q/H/Q) required refractive index orderings of $n_s < n_{f1} < n_{f2}$ and $n_{f2} > n_{f3} > n_a$ for antireflection properties. The middle layer (n_{f2}) acts as an absentee layer when the incident wavelength is the same as the reference wavelength ($\lambda = \lambda_0$). The reflection coefficient is equal to:

$$r = \frac{n_a - n_{f3}}{n_a + n_{f3}} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f2}^2}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f2}^2}\right)}. \quad (14)$$

Because the middle film layer (n_{f2}) is HWOT thick, it behaves as an absentee layer [23, 27]. So $n_{f2} = n_{f1}$, and Eq. 14 becomes:

$$r = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}^2}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}^2}\right)} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}, \quad (15)$$

a three-layer antireflection condition occurs when:

$$\frac{n_a}{n_s} = \frac{n_{f3}^2}{n_{f1}^2}. \quad (16)$$

The refractive index ratio of top and bottom films is similar to that of the two-layer design. As an example, if the bottom layer (n_{f1}) is Sb₂Se₃ (QWOT), then the ideal top layer ($n_{f3, ideal}$) is 2.78, so TiO₂ ($n_{TiO2} = 2.31$) is used again. The middle layer needs to have the highest refractive index, so Silicon (Si) is used ($n_{Si} = n_{f2} = 3.55$) [28]. As shown in Fig.3, the reflectance of the three-layer design (yellow) at wavelengths further from λ_0 (i.e., 1400 nm and longer) is lower than that of a single (blue) or two-layer design (green). This is a benefit of adding an extra absentee layer in the middle, it contributes to a broader operating bandwidth [27].

Instead of placing the PCM as the bottom film layer, Sb₂Se₃ can be moved to the middle so that the selection of ideals n_{f1} and n_{f3} is easier. The bottom layer could be Si₃N₄ ($n_{f1} = 2.00$) [29] and the ideal top layer calculated using Eq. 16 has $n_{f3} = 1.65$, which Al₂O₃ has the closest refractive index ($n_{Al2O3} = 1.62$) [26]. These designs are shown in Fig. 3, where the reflectance of the designs with Sb₂Se₃ deposited directly on top of the substrate, i.e., single-layer, two-layer, and three-layer designs, are ~6.5%. But the three-layer design with Sb₂Se₃ placed in the middle ($n_{f2} = \text{Sb}_2\text{Se}_3$) has a reflectance of 3.4%, which is the same value as that of an ideal single-layer design (3.4%). This improvement can be explained by the selection of n_{f1} and n_{f3} which are closer to ideal values.

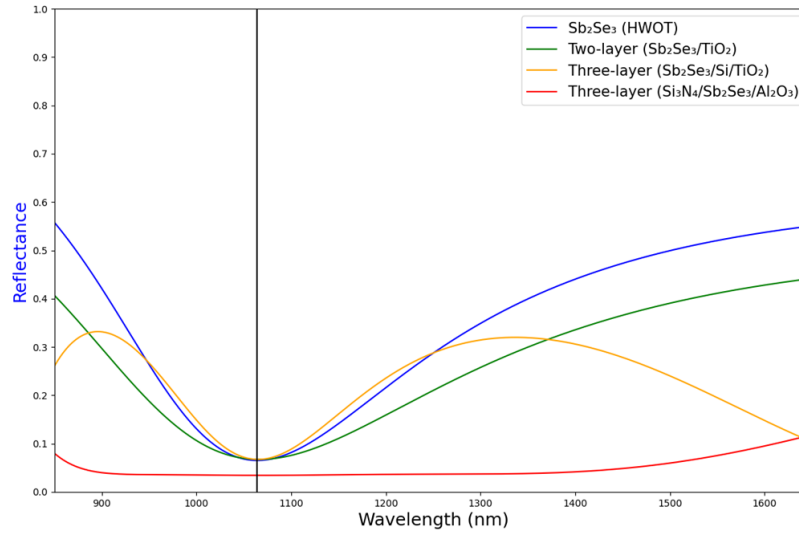


Fig. 3. Reflectance comparison between a single-layer and multi-layer Sb_2Se_3 designs. At 1064nm (black vertical line), the single-layer, two-layer and three-layer designs with Sb_2Se_3 as bottom layer (n_H) have similar reflectance at 6.5 - 6.7%. The three-layer design with Sb_2Se_3 as middle layer (n_D) has the lowest reflectance at 3.4%.

2.4 Long-pass edge-filter design

In all previous examples, the reflectance after Sb_2Se_3 transitioned to the crystalline phase have not yet been considered. In addition to low reflectance in the ON mode, another requirement for transmittance tuning is a high optical contrast between the two modes. Figure 4 shows a reflectance comparison between Sb_2Se_3 in amorphous (solid) and crystalline (dotted) phases, which represent the devices in the ON and OFF modes, respectively. These designs are the same as those described in Fig. 3. The single-layer design exhibited the highest reflectance contrast of 55.0% (Fig. 4a, blue). But for the three-layer design with Sb_2Se_3 as the middle layer (Fig. 5b, red), even though it has the lowest reflectance in amorphous phase (3.4%), its crystalline phase reflectance only increased to 14.8%, which results in a reflectance contrast of only 11.4%.

Another antireflection design is called an edge-filter design, which has high reflection in the region surrounding the reference wavelength (λ_0) but exhibits strong antireflection (AR) properties outside of this region. The filter can be designed such that the desired operating wavelength is in the AR region, at the edge of these two regions. When phase transition occurs and the device changes to OFF mode, the reflectance profile will be shifted and the operating wavelength will be inside the high-reflection region instead. The edge filter design is composed of QWOT-thick high refractive index (H) and low refractive index (L) layers deposited symmetrically on top of each other in a structure called unit cell (U). Only one side of the AR region have a stable reflectance profile, determined by the unit cell structure; long-pass structure ($\frac{H}{2}L\frac{H}{2}$) has a stable AR at wavelengths longer than the high-reflection region, but the short-pass structure ($\frac{L}{2}H\frac{L}{2}$) has a stable AR at shorter wavelengths, the $1/2$ denotes a layer with half-QWOT thickness. The number of unit cells could be one or more, and this number is directly correlated to the steepness of regions' edges. To minimize the thickness of the device, a long-pass design is preferred because λ_0 is at a shorter wavelength, so optical and physical thickness will be thinner. The spectral width of the high-reflection region is directly correlated with the differences between the H and L refractive indices.

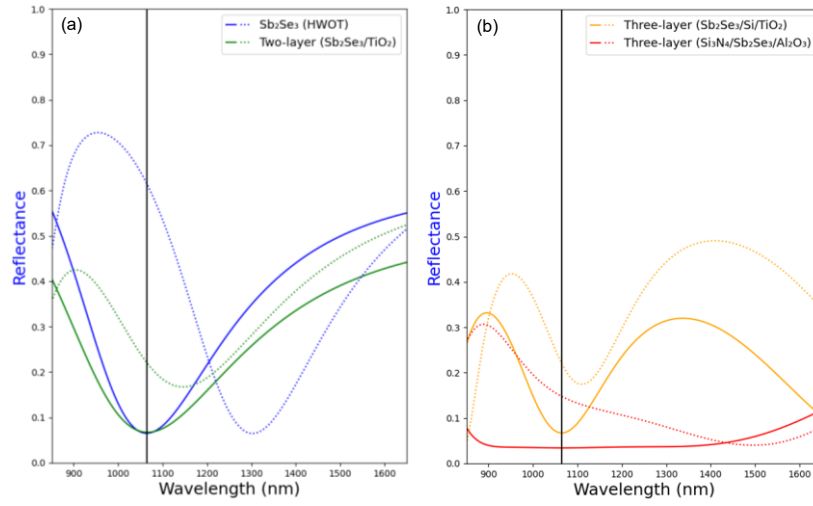


Fig. 4. Reflectance of the same designs shown in Fig. 3 before (solid) and after (dotted) phase transition. (a) Single-layer (blue) and two-layer design (green). (b) Three-layer designs with Sb₂Se₃ as the bottom layer (orange) and Sb₂Se₃ as middle layer (red). Single-layer design has the highest reflectance contrast. The vertical black lines indicate $\lambda = 1064$ nm.

Figure 5 shows five long-pass edge filter designs with number of unit cells (N) from 1 to 8. These designs used Sb₂Se₃ as the H layer and MgF₂ as the L layer; the reference wavelength is changed to $\lambda_0 = 740$ nm. When these devices are in the amorphous phase (Fig. 5a), the wavelengths surrounding the operating wavelength ($\lambda = 1064$ nm) are in the AR region, when they are switched to crystalline phase (Fig. 5b) the wavelength of interest is in the high-reflection region. These shifts cause the reflectance contrast to be higher than any previous designs, with more than 90% contrast for four, six and eight unit-cell designs. It can also be seen that increasing N makes the region's edge steeper, but the change from 6 to 8 unit cells is lower than from 1 to 2 unit cells.

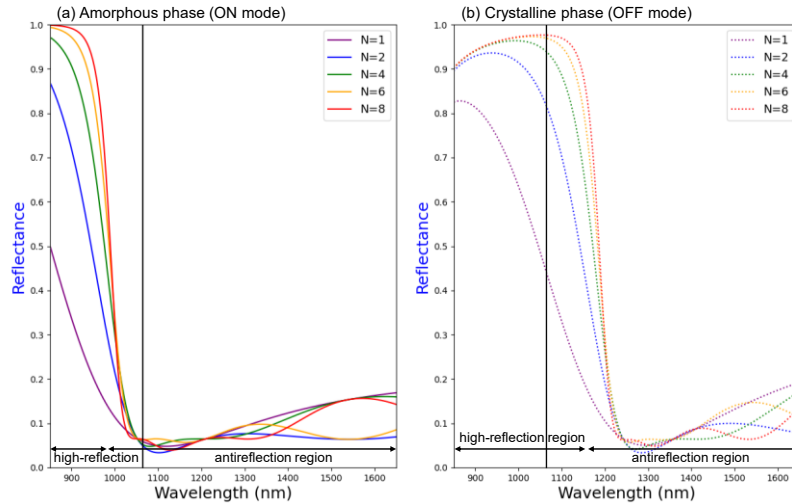


Fig. 5. Reflectance comparison of long-pass edge-filter designs with increasing number of unit cells (N). (a) When devices are in amorphous phase (ON mode), the interested wavelength (black vertical line) is in AR region. (b) When devices are in crystalline phase (OFF mode), the interested wavelength is in high-reflection region.

3. Methods

3.1 Simulation of optical performances

In this study, we use the transfer matrix method (TMM) of Fresnel equation to calculate the transmittance, absorptance, and reflectance of each design, it is based on [23], although another simulation software such as COMSOL, or OpenFilters [30] can also be used. We use 3-mm thick silica (SiO_2) as the substrate ($n_s = 1.47$) [24]. The high-index layers (H) are Si [28], TiO_2 [26], and Si_3N_4 [29], their refractive indices at 1064 nm are 3.56, 2.31, and 2.00, respectively. The low-index layers (L) are Al_2O_3 [26], SiO_2 and MgF_2 [25], and their refractive indices are 1.62, 1.45, and 1.38, respectively. All the aforementioned refractive index datasets can be found on *refractiveindex.info* website [31]. However, because antimony triselenide (Sb_2Se_3) is a relatively new material in this field, its dataset cannot be found there, we instead used the Sb_2Se_3 dataset from Delaney, Zeimpekis, Lawson, Hewak and Muskens [18]. The refractive index of air is set to $n_a = 1.0$ across the spectrum, and the backside reflection from the substrate-air layer is calculated. The incident angle is 0° , and light is transverse electric (TE) polarized unless stated otherwise.

All the designs are intended to operate at a wavelength center around 1064 nm. The design comparison is between single-layer, two-layer, three-layer, and edge-filter designs. The goal is to design a device with a high transmission contrast (ΔT) between the two phases of Sb_2Se_3 by minimizing the reflectance in the ON mode and maximizing R in the OFF mode.

3.2 Simulation of optical endurance and stability under large phase switching cycles

Experimental studies have demonstrated that Sb_2Se_3 can has phase switching cycles with minimal degradation of its optical contrast and maintain a low-loss, reversible modulation in the order of 10^6 phase switching cycles [21]. Even so, Sb_2Se_3 still exhibits a slight complex refractive index drift due to structural relaxation and cumulative thermal damage. Accounting for these variations is essential for ensuring long-term reliability and precision in high-endurance photonic applications [21, 22]. To account for this, the complex refractive index ($\tilde{n}$) can be rewritten as:

$$\tilde{n}_p(C) = n_p(C) + ik_p(C), \quad (17)$$

where C is the number of phase switching cycle. The real part of the refractive index (n) is described by an exponential relaxation towards a saturated value such that:

$$n_p(C) = n_{p,\infty} - (n_{p,\infty} - n_{p,0})\exp^{-\alpha_p C}, \quad (18)$$

where $n_{p,\infty}$ is the asymptotic refractive index after many cycles, $n_{p,0}$ is the refractive index at the first cycle (obtained from ellipsometry or literature data), and α_p is the drift coefficient per cycle of phase p [18, 32].

Meanwhile, the imaginary part (k) can be approximated from a linear growth as:

$$k_p(C) = k_{p,0}(1 + \beta_p C), \quad (19)$$

where $k_{p,0}$ is the initial loss, and β_p is the loss growth coefficient. Parameter β_p is the relative loss per cycle of phase p [21, 33]. The model from Eq. (19) is adopted in this work, they are consistent with experimental trends and with general PCM endurance theory, providing a practical way to assess contrast degradation and threshold drift of PCM-based devices over large cycles [18, 21, 32, 33].

Optical endurance can be quantified using the following metrics; percentage transmitting contrast degradation (D_{pct}) and insertion loss (IL). Transmittance contrast degradation is defined as the change in transmittance contrast after N cycles:

$$D_{pct}(C) = \frac{\Delta T(C) - \Delta T_0}{\Delta T_0}, \quad (20)$$

where ΔT_0 is the transmittance contrast between the amorphous and crystalline phases in the initial cycle $C = 0$ ($\Delta T_0 = T_a(0) - T_c(0)$). A negative D_{pct} indicates a decreasing contrast.

The insertion loss (IL) of the device at cycle C compares the high-transmittance amorphous phase after C cycle to the initial cycle:

$$IL_{dB}(C) = 10 \log \left[\frac{T_a(C)}{T_a(0)} \right]. \quad (21)$$

A growing $IL(C)$ indicates increasing loss, which is typically driven by the linear growth of the imaginary index $k_p(C)$ and by detuning of the resonance due to drift in $n_p(C)$.

4. Results & Discussions

4.1 Optical properties of 1064-nm optical filters

The design architectures, number of layers, total thickness of Sb_2Se_3 , overall design thickness, transmittance in each mode, and transmittance contrast ($\Delta T = T_a - T_c$) of each design are shown in Table 1. The operating wavelength is 1064 nm, and the reference wavelength (λ_0) is 1064 nm for single-layer, two-layer, and three-layer designs; however, λ_0 is 740 nm for edge-filter designs. The results show that only single-layer and edge-filter designs with at least two unit cells ($N \geq 2$) have a ΔT higher than 50%, with 7-unit-cells design having the highest ΔT of 95.1%. However, there is only marginal improvement in ΔT when N is increased above 3, but the Sb_2Se_3 required and the overall device size increased drastically; for example, ΔT improved by 2% in $N = 4$ design when compared to $N = 3$ design, the thickness increased by 33.3%, but when comparing $N = 5$ to $N = 4$ designs, ΔT increased by only 0.6%, but the device needs to be 25.0% thicker. It is important to note that increasing unit cell numbers also require more Sb_2Se_3 material, less Sb_2Se_3 used also has the benefit of minimizing the energy required for phase transition.

Table 1. Designs of 1064-nm Sb_2Se_3 -based optical filter. Transmittance is calculated at the interested $\lambda = 1064$ nm. Single, two, and three-layer designs all have $\lambda_0 = 1064$ nm, but all edge-filter designs have $\lambda_0 = 740$ nm.

Design	Layout	Number of layers	SbSe used (nm)	Device size (nm)	T (ON) (%)	T (OFF) (%)	ΔT (%)
Single-layer	SbSe	1	157.9	157.9	93.5	37.2	56.3
Two-layers	SbSe/TiO	2	78.9	194.1	93.3	76.9	16.4
Three-layers	SbSe/MgF/TiO	3	78.9	580.3	93.3	76.9	16.4
Three-layers	SiN/SbSe/AlO	3	157.9	455.7	96.6	83.3	13.3
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^1$	3	48.5	182.5	94.1	54.3	39.8
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^2$	5	97.0	365.0	95.0	16.3	78.7
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^3$	7	145.5	547.6	94.8	4.0	90.8
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^4$	9	194.0	730.1	93.7	0.9	92.8
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^5$	11	242.6	912.6	93.6	0.2	93.4
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^6$	13	291.1	1,095.1	94.5	0.0	94.5
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^7$	15	339.6	1,277.7	95.1	0.0	95.1
Edge filter	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^8$	17	388.1	1,460.2	94.4	0.0	94.4

In the edge-filter designs, the transmittance in the OFF mode decreased with increasing N (Fig. 5b), with $N \geq 6$ designs having zero transmittance. This is because N correlates to the total Sb_2Se_3 thickness and Sb_2Se_3 in the crystalline phase has a small, but not zero, extinction coefficient ($k_{\text{cry}}(1064) = 0.0097$), therefore, light is absorbed by each Sb_2Se_3 layer. For reference, $k_{\text{Sb}_2\text{Se}_3}$ is in an order of 10^2 smaller than that of other PCMs such as GST ($k_{\text{GST, cry}} = 2.4540$) or VO_2 ($k_{\text{VO}_2, \text{metallic}} = 1.4580$). These are illustrated in Fig. 6, the transmittance contrast (ΔT) of the single-layer and edge-filter designs with $N \leq 6$ are shown. Even though edge-filter designs with N from 3 to 6 (cyan, green, yellow, and orange, in order) have similar ΔT at the operating wavelength (black vertical line), the bandwidth of high- ΔT region is wider in design with larger unit-cell numbers. This property should be considered if the filter is intended to be used in broad spectrum.

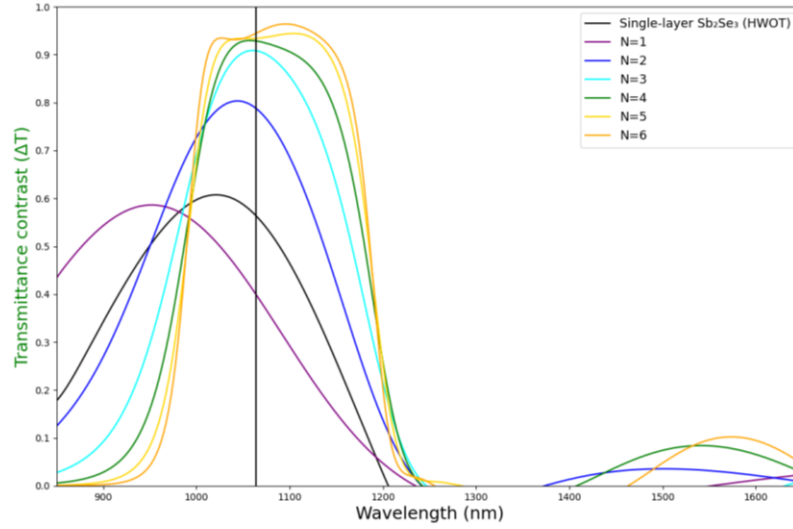


Fig. 6. Transmittance contrast (ΔT) of a single-layer and edge-filter designs with different number of unit cells (N). The vertical black line indicates the interested 1064 nm wavelength.

Two other properties that might need to be considered are the optical performance at varying incident angles and the polarization mode of the incident light (TE or TM). Figure 7 shows 4-unit-cells and 5-unit-cells designs with incident angles ranging from 0° to 90° , and with polarization in TE (Fig. 7a and 7c) and TM (Fig. 7b and 7d). For both TE and TM polarized light, ΔT remains higher than 90% from 0° to $\sim 30^\circ$, but TM polarization causes the high- ΔT region to blueshift significantly at higher incident angle. The spectral widths of the high- ΔT region in both designs are comparable, but the 5-unit-cells design has a higher ΔT across the spectrum, which indicates a better broadband, wide-angle, tunable optical device.

With the reasons given above, we decided that 5-unit-cells, long-pass edge-filter design (Fig. 6, yellow) has the best compromise between high optical performances, device size, and complexity. This design has a broadband transmittance contrast between the amorphous (ON mode) and the crystalline phases (OFF mode), and the FWHM of high- ΔT region is 197.0 nm. This design is used as an example in the following sections.

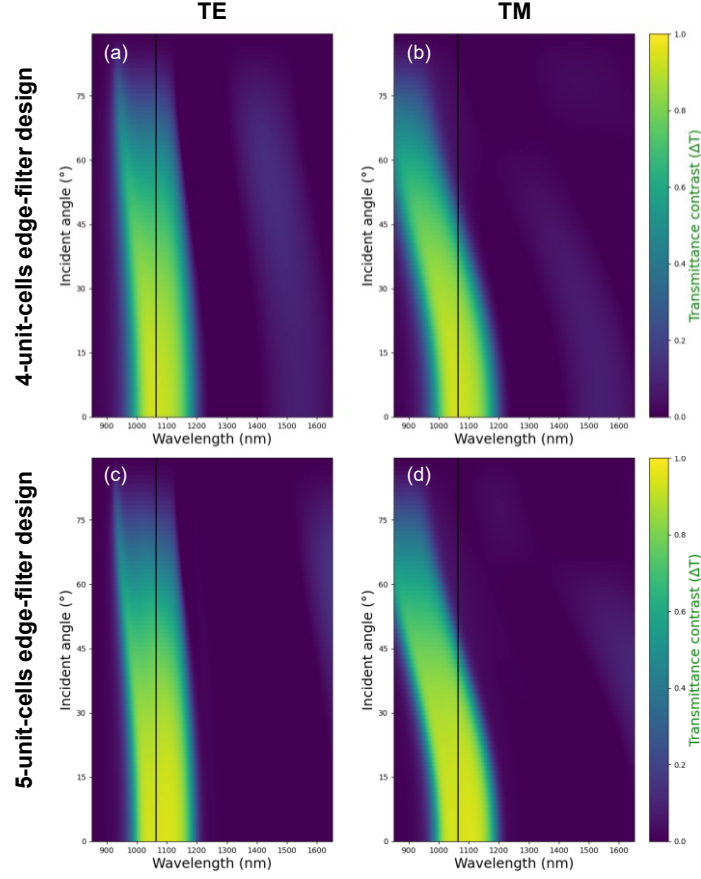


Fig. 7. Transmittance contrast at different incident angles and wavelengths of 4-unit-cells and 5-unit-cells edge-filter designs. The polarizations of incident light are TE (a, c) and TM (b, d).

4.2 Endurance simulation of 1064-nm optical filters(check endurance of edge-filter.)

Endurance simulation at 1064 nm of the single layer Sb_2Se_3 design, 4-unit-cells, and 5-unit-cells edge-filter designs demonstrated that all three resonant optical filters remain operational up to 1.5×10^6 phase switching cycles, with design-dependent degradation of their transmittance metrics. For the single-layer design, both the amorphous (T_a) and crystalline phases transmittance (T_c) decrease modestly (Fig. 8a). And both the corresponding percentage transmittance contrast degradation (D_{pct}) and insertion loss (IL) also decrease with cycling number, Fig. 8d and 8g, respectively. In the 4-unit-cells edge-filter design, the T_a and insertion loss decreased less than in the single layer design, but its T_c increased slightly. This cause the contrast degradation to become two times larger (Fig. 8e). The 5-unit-cells edge-filter design further accentuates these trends: its T_a and IL decrease more than 4-unit-cells design, but still less than a single-layer design. While its D_{pct} is now three times larger than the single-layer design (Fig. 8f).

These results confirm that designs with multiple layers of Sb_2Se_3 have comparable transmittance endurance to a single-layer Sb_2Se_3 . The edge-filter designs have a stronger field confinement so they exhibit less insertion loss when compared to single-layer design, their larger D_{pct} are attributed to fact that their T_c become higher with cycle numbers. Overall, the edge-filter designs still have a higher transmittance contrast and a lower insertion loss after 1.5×10^6 cycles.

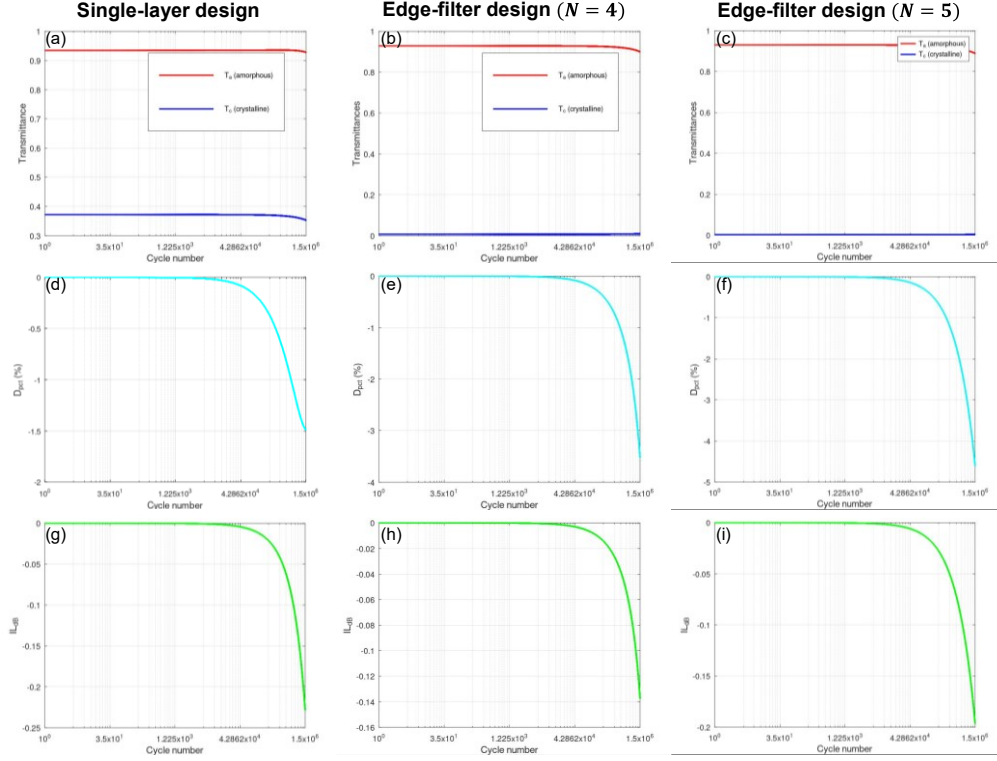


Fig. 8. Optical endurance at 1064 nm of the single-layer, 4-unit-cells edge-filter, and 5-unit-cells edge-filter designs after 1.5×10^6 phase-switching cycles. (a-c) Transmittance in amorphous (red) and crystalline phases (blue). (d-f) The percentage transmittance contrast degradation relative to the initial value. (g-i) Insertion loss in amorphous phase, expressed in dB. Not a difference y-axis scale.

4.3 Transmittance tunability and optimization.

Edge-filter designs have multiple layers of Sb_2Se_3 which complicates the phase switching setup. When laser-induced heating is used as a switching method, the phase transition of every Sb_2Se_3 layer does not occur simultaneously and the transmission changes gradually. This effect is shown in the optical limiter of Sarangan, Duran, Vasilyev, Limberopoulos, Vitebskiy and Anisimov [3]. Their device has a similar long-pass edge-filter design $(\frac{H}{2} L \frac{H}{2})^4$ but the H layers are GST instead of Sb_2Se_3 , and the L layers are SiO_2 instead of MgF_2 . Their experimental results showed that the phase switching of GST layers occurs layer-by-layer in an order that is determined by the changes in the optical field distribution and thermal crosstalk between layers, which does not necessarily start from the air -side to the substrate-side. However, our device is intended to be used as a transmittance-tunable filter that does not require a fast switching speed; therefore, electrical heating is also available as an alternative phase switching method. The ability to manually control the phase of each Sb_2Se_3 layer enables the filter to have multiple transmittance values. Using the 5-unit-cells edge-filter design described in Table 1, which consists of 11 layers of Sb_2Se_3 and MgF_2 (six layers of Sb_2Se_3), there would be a total of 2^6 configurations for the Sb_2Se_3 phases. Figure 9 highlights some of these configurations; the legend denotes Sb_2Se_3 in the amorphous (a) or crystalline phase (c), and the ordering from left to right is from the substrate to the air side (f_1 to f_6). With the correct phase configuration, the transmittance can be gradually reduced from peak of 93.4% to 0.0%.

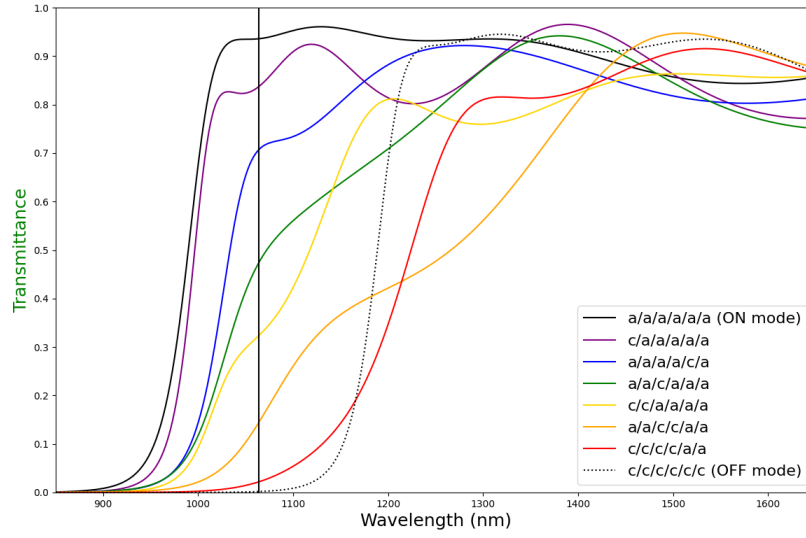


Fig. 9. Transmittance of the five-unit-cells edge-filter design in 8 different configurations. Each Sb_2Se_3 layer is in amorphous (a) or crystalline (c) phases, and the layout ordering from left to right is substrate-side to air-side.

The peak transmittance contrast of each design shown in Fig. 6 does not always correspond to the desired operating wavelength of 1064 nm. This is explained by the arbitrary selection of the reference wavelength in edge-filter designs. So, an optimization algorithm can be used to improve the optical performance. A simple merit function found in OpenFilters [30] is used in this example. The optimization goal for the 5-unit-cells edge-filter design is to have a transmittance value of at least 97% from 1000 to 1200 nm at intervals of every 20 nm. The optimization took less than 5 s, and the new design now required 304.2 nm of Sb_2Se_3 , 25% more than the original design, but the overall thickness is now 845.6 nm, 7% less than the original, the thickness of each layer is shown in Table 2. Figure 10a shows the transmittance of the original (green) and optimized designs (blue) in the amorphous (solid) and crystalline phases (dotted). Overall, the optimized design has a higher transmittance in the desired 1000 to 1200 nm spectrum, but the spectral profile has been blue shifted by a small amount. The transmittance contrast at 1064 nm (Fig. 10b) of the optimized design is also higher, 95.8%, compared to 93.4% of the original. The transmittance contrast FWHM also extended slightly to 207.4 nm (original: 197.0 nm).

Table 2. Thickness of each layer before and after optimization. Sb_2Se_3 and MgF_2 layers units are in nm.

Design	SiO_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3
Original	3 mm	24.3	134.0	48.5	134.0	48.5	134.0	48.5	134.0	48.5	134.0	24.3
Optimized	3 mm	46.1	67.5	59.3	148.1	55.0	89.3	50.0	145.3	63.3	91.2	30.5

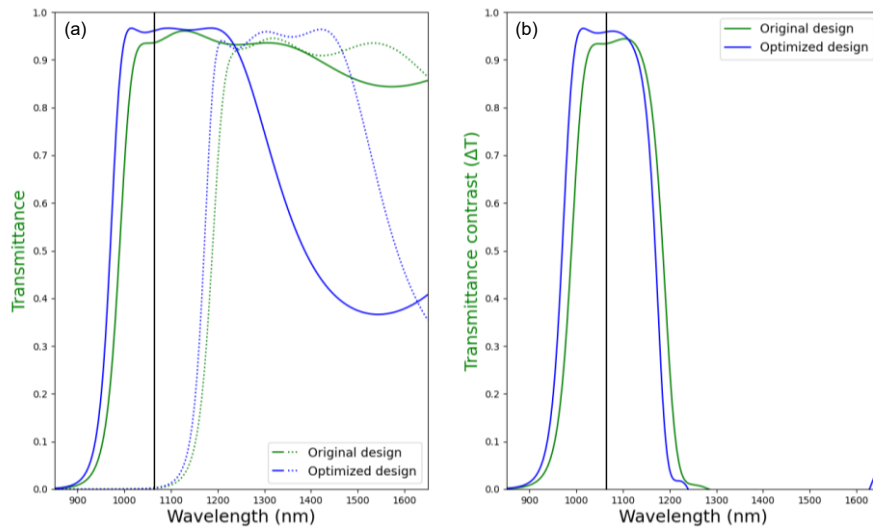


Fig. 10. Transmittance comparison of 5-unit-cell edge-filter design before (green) and after (blue) optimization. (a) The transmittance in ON mode (solid) and OFF mode (dotted). (b) The transmittance contrast (ΔT). An optimized design has higher optical performance from 900 to 1200 nm, but the high- ΔT region has been blueshifted by a small amount.

5. Conclusions

We have shown that Sb_2Se_3 is a suitable PCM for high-efficient, broadband transmittance-tunable optical filter operating around 1064 nm wavelength. The 5-unit-cells long-pass edge-filter design has the best compromise between optical performance, size, and design complexity. This design required six Sb_2Se_3 layers and five MgF_2 layers and is 912 nm thick. Transmittance at 1064 nm in the ON mode is 93.4% and 0.2% in OFF mode, a contrast of 93.4% with around 200 nm FWHM. Simulations show that after 1.5×10^6 phase switching cycles, the transmittance contrast degradation and insertion loss in ON mode is still within acceptable ranges. The multilayer nature of edge-filter design enables the gradual modulation of transmittance by changing the phase of each Sb_2Se_3 layer. We also show that a quick and simple optimization algorithm can increase the optical performance from 900-1200 nm.

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Data availability. Data underlying the results presented in this paper may be obtained from the corresponding author upon a reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

The authors used Gemini for Python codes debugging, Gemini and Paperpal Preflight to check the grammar of the manuscript, and Consensus was used to help with the initial literature review. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the contents published.

References

1. K. Matuszek, M. Kar, J. M. Pringle et al., "Phase Change Materials for Renewable Energy Storage at Intermediate Temperatures," *Chemical Reviews* **123**, 491–514 (2023).
2. R. Matos, and N. Pala, "A Review of Phase-Change Materials and Their Potential for Reconfigurable Intelligent Surfaces," *Micromachines* **14**, 1259 (2023).
3. A. Sarangan, J. Duran, V. Vasilyev et al., "Broadband Reflective Optical Limiter Using GST Phase Change Material," *IEEE Photonics Journal* **10**, 1–9 (2018).
4. K. Aryana, H. J. Kim, M. R. Islam et al., "Optical and thermal properties of $\text{Ge}_2\text{Sb}_2\text{Te}_5$, Sb_2Se_3 , and Sb_2S_3 for reconfigurable photonic devices [Invited]," *Opt. Mater. Express* **13** (2023).

5. W. Jia, R. Menon, and B. Sensale-Rodriguez, "Unique prospects of phase change material Sb₂Se₃ for ultra-compact reconfigurable nanophotonic devices," *Opt. Mater. Express* **11**, 3007–3014 (2021).
6. S. Blundell, T. W. Radford, I. A. Ajia et al., "Ultracompact Programmable Silicon Photonics Using Layers of Low-Loss Phase-Change Material Sb₂Se₃ of Increasing Thickness," *ACS Photonics* **12**, 1382–1391 (2025).
7. C. Wan, Z. Zhang, J. Salman et al., "Ultrathin Broadband Reflective Optical Limiter," *Laser & Photonics Reviews* **15**, 2100001 (2021).
8. A. Tognazzi, M. Gandolfi, B. Li et al., "Opto-thermal dynamics of thin-film optical limiters based on the VO₂ phase transition," *Opt. Mater. Express* **13**, 41–52 (2023).
9. Y. Li, C. Zhou, Y. Feng et al., "An Adaptive Optical Limiter Based on a VO₂/GaN Thin Film for Infrared Lasers," *Photonics* **13**, 148 (2026).
10. Y. Zhang, J. B. Chou, J. Li et al., "Broadband transparent optical phase change materials for high-performance nonvolatile photonics," *Nature Communications* **10**, 4279 (2019).
11. H. Li, D. Zeng, Y. Huang et al., "Ultra-compact, ultra-low-loss and broadband 2 x 2 nonvolatile optical switch based on Sb(2)Se(3) phase change material," *Opt Express* **32**, 35287–35297 (2024).
12. R. Heenkenda, K. Hirakawa, and A. Sarangan, "Tunable optical filter using phase change materials for smart IR night vision applications," *Opt. Express* **29**, 33795–33803 (2021).
13. D. d. Ceglie, M. Grande, A. Vincenti et al., "Tunable Filters for Visible Light Based on Resonant VO₂ Planar Thin Films," in *2020 22nd International Conference on Transparent Optical Networks (ICTON)*(2020), pp. 1–4.
14. Y.-S. Huang, C.-Y. Lee, M. Rath et al., "Tunable structural transmissive color in fano-resonant optical coatings employing phase-change materials," *Materials Today Advances* **18**, 100364 (2023).
15. M. Cui, Y. Xie, P. Yang et al., "Tunable multi-channel optical filter based on the near-infrared ultralow-loss phase change material Sb₂Se₃," *Opt. Express* **34**, 7378–7387 (2026).
16. X. Tian, and Z.-Y. Li, "An Optically-Triggered Switchable Mid-Infrared Perfect Absorber Based on Phase-Change Material of Vanadium Dioxide," *Plasmonics* **13**, 1393–1402 (2018).
17. C. Liu, Y. Yuan, L. Cheng et al., "Tunable nonlinear optical absorption in amorphous and crystalline Sb₂Se₃ thin films," *Journal of Alloys and Compounds* **791**, 753–760 (2019).
18. M. Delaney, I. Zeimpekis, D. Lawson et al., "A New Family of Ultralow Loss Reversible Phase-Change Materials for Photonic Integrated Circuits: Sb₂S₃ and Sb₂Se₃," *Advanced Functional Materials* **30** (2020).
19. A. A. Burtsev, N. N. Eliseev, V. A. Mikhalevsky et al., "Physical properties' temperature dynamics of GeTe, Ge₂Sb₂Te₅ and Ge₂Sb₂Se₄Te₁ phase change materials," *Materials Science in Semiconductor Processing* **150**, 106907 (2022).
20. X. Chen, J. Lin, and K. Wang, "A Review of Integrated Photonic Devices Using Sb₂Se₃," *Advanced Physics Research* **4**, 2400080 (2024).
21. D. Lawson, S. Blundell, M. Ebert et al., "Optical switching beyond a million cycles of low-loss phase change material Sb₂Se₃," *Opt. Mater. Express* **14** (2023).
22. M. S. Alam, R. Laing, Z. Bolatbek et al., "Fast Cycling Speed with Multimillion Cycling Endurance of Ultra-Low Loss Phase Change Material (Sb₂Se₃) by Engineered Laser Pulse Irradiation," *Advanced Functional Materials* **34**, 2310306 (2024).
23. A. Sarangan, *Optical Thin Film Design* (CRC Press, 2020).
24. D. Franta, D. Nečas, I. Ohlídal et al., *Optical characterization of SiO₂ thin films using universal dispersion model over wide spectral range* (SPIE, 2016).
25. D. Franta, D. Nečas, A. Giglia et al., "Universal dispersion model for characterization of optical thin films over wide spectral range: Application to magnesium fluoride," *Applied Surface Science* **421**, 424–429 (2017).
26. S. V. Zhukovsky, A. Andryieuski, O. Takayama et al., "Experimental Demonstration of Effective Medium Approximation Breakdown in Deeply Subwavelength All-Dielectric Multilayers," *Physical Review Letters* **115**, 177402 (2015).
27. Y. Li, K. Yang, B. Xia et al., "Preparation of mechanically stable triple-layer interference broadband antireflective coatings with self-cleaning property by sol–gel technique," *RSC Advances* **7**, 14660–14668 (2017).
28. D. Franta, P. Franta, J. Vohánka et al., "Determination of thicknesses and temperatures of crystalline silicon wafers from optical measurements in the far infrared region," *Journal of Applied Physics* **123** (2018).
29. L. Y. Beliaev, E. Shkondin, A. V. Lavrinenko et al., "Optical, structural and composition properties of silicon nitride films deposited by reactive radio-frequency sputtering, low pressure and plasma-enhanced chemical vapor deposition," *Thin Solid Films* **763**, 139568 (2022).
30. S. Larouche, and L. Martinu, "OpenFilters: open-source software for the design, optimization, and synthesis of optical filters," *Appl. Opt.* **47**, C219–C230 (2008).
31. M. N. Polyanskiy, "Refractiveindex.info database of optical constants," *Scientific Data* **11**, 94 (2024).
32. L. Martin-Monier, C. C. Popescu, L. Ranno et al., "Endurance of chalcogenide optical phase change materials: a review," *Opt. Mater. Express* **12**, 2145–2167 (2022).
33. Y. Zhang, C. Rios, M. Y. Shalaginov et al., "Myths and truths about optical phase change materials: A perspective," *Applied Physics Letters* **118** (2021).